

BISHKEK, Kyrgyzstan

WMO: 383530

Lat: 42.850N

Lon: 74.533E

Elev: 760

StdP: 92.52

Time Zone: 6.00 (E06)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.8	-19.6	0.7	-16.2	-16.8	0.9	-12.9	7.3	3.4	5.7	3.4	1.0	180	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	35.5	18.7	34.0	18.1	32.6	17.7	21.4	31.7	19.9	30.0	18.8	29.2	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	14.0	27.1	16.1	12.6	24.6	15.0	11.7	23.9	66.1	31.5	60.5	30.0	56.5	29.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	5.9	4.6	-19.0	38.4	4.3	1.6	-22.2	39.5	-24.7	40.4	-27.1	41.3	-30.3	42.5		
			-19.4	21.3	4.2	2.8	-22.4	23.4	-24.9	25.0	-27.2	26.6	-30.3	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-1.8	0.2	7.3	13.1	18.2	23.1	25.3	24.4	19.0	12.1	5.1	-0.7
	DBStd	10.67	5.63	5.79	5.54	4.72	3.74	3.34	2.89	3.05	4.15	4.85	5.41	5.69
	HDD10.0	1314	367	277	118	24	1	0	0	0	1	34	161	332
	HDD18.3	2923	625	509	343	165	48	5	0	1	43	199	396	590
	CDD10.0	2104	0	2	34	116	256	392	475	446	270	99	16	1
	CDD18.3	672	0	0	1	7	44	147	217	188	62	6	1	0
	CDH23.3	7633	0	0	12	103	464	1604	2566	2150	653	74	6	0
	CDH26.7	3327	0	0	2	20	133	684	1250	994	227	17	1	0
Wind	WSAvg	1.7	1.4	1.6	1.9	2.0	2.0	2.1	2.0	1.9	1.7	1.6	1.4	1.4
Precipitation	PrecAvg	439	25	32	46	74	66	36	19	12	17	39	43	30
	PrecMax	671	58	84	104	157	157	96	87	51	56	102	118	84
	PrecMin	290	5	7	15	12	24	4	0	0	0	5	8	4
	PrecStd	84	12	15	20	34	32	23	20	11	16	23	24	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.3	25.2	29.0	32.8	37.0	38.0	36.8	33.7	29.1	23.6	14.4
		MCWB	6.5	8.5	13.1	15.0	19.5	19.4	18.7	18.8	17.5	14.5	11.9	6.3
	2%	DB	10.0	13.0	21.5	26.1	29.8	34.1	35.9	35.0	31.4	25.2	18.3	11.1
		MCWB	4.3	6.6	11.1	14.3	17.3	18.2	18.7	18.2	16.3	12.8	9.7	4.8
	5%	DB	7.5	10.2	18.6	23.8	27.9	32.3	34.2	33.4	29.3	22.7	15.3	8.6
		MCWB	2.8	5.0	10.0	13.5	16.5	17.9	18.1	17.8	15.7	11.8	8.3	3.6
	10%	DB	5.1	7.8	16.0	21.4	25.9	30.6	32.7	31.8	27.2	20.1	12.7	6.4
		MCWB	1.6	3.9	9.0	12.6	15.6	17.2	17.6	17.5	14.8	10.8	6.9	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	9.1	14.5	17.3	22.0	23.8	23.7	23.1	19.8	15.7	12.8	7.4
		MCDB	12.3	16.1	22.9	25.7	31.5	33.6	34.7	33.4	30.9	25.4	20.3	12.1
	2%	WB	4.8	7.0	12.1	15.8	18.7	20.6	21.0	20.7	17.7	13.6	11.0	5.4
		MCDB	9.3	12.0	18.7	23.7	27.4	30.3	30.8	31.5	28.6	22.6	17.8	10.1
	5%	WB	3.1	5.5	10.6	14.5	17.3	19.0	19.5	19.3	16.3	12.6	8.8	4.0
		MCDB	6.7	9.5	17.0	21.2	25.4	28.9	29.7	30.2	26.8	20.6	13.7	8.1
	10%	WB	1.8	4.0	9.4	13.3	16.2	17.9	18.5	18.2	15.1	11.6	7.4	2.7
		MCDB	4.7	7.8	15.4	19.8	24.3	27.8	29.3	29.3	26.0	18.7	12.0	6.0
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	10.3	11.0	11.9	12.5	12.7	13.0	12.6	11.3	9.3	8.9
		MCDBR	12.2	12.5	14.9	15.4	14.9	15.2	14.6	14.8	15.2	15.1	13.6	12.4
		MCWBR	7.7	7.5	7.5	7.1	6.9	6.2	5.1	5.6	6.4	6.9	7.2	7.6
	5% WB	MCDBR	10.9	11.7	13.6	13.3	13.7	14.3	13.8	14.1	14.6	13.2	12.0	11.1
		MCWBR	7.2	7.4	7.3	7.1	7.3	7.2	6.5	7.0	7.1	6.5	6.8	7.2
Clear-Sky Solar Irradiance	taub	0.335	0.381	0.451	0.482	0.455	0.459	0.451	0.418	0.401	0.419	0.384	0.329	
	taud	2.134	2.031	1.908	1.900	2.046	2.082	2.119	2.203	2.187	2.034	2.061	2.144	
	Ebn at Noon	767	787	774	787	823	820	822	838	821	734	692	732	
	Edh at Noon	101	133	171	186	165	160	153	136	128	131	109	91	
All-Sky Solar Radiation	RadAvg	2.09	3.03	4.01	4.96	6.08	6.68	6.75	6.31	5.02	3.14	2.02	1.69	
	RadStd	0.15	0.23	0.34	0.47	0.46	0.36	0.39	0.34	0.26	0.38	0.22	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	-1.93	-2.98	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.84	N/A	N/A	N/A	N/A

Nomenclature: See separate page